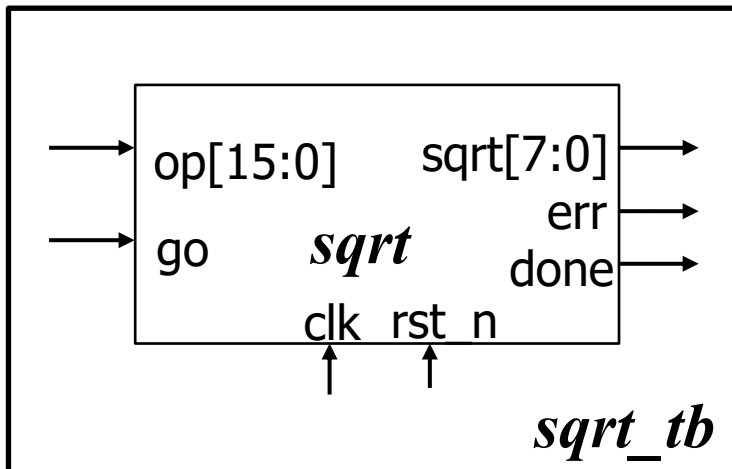




Exercise 07: Sequential Circuit Testbench

- A sequential DUT (**sqrt.sv**) is provided.
- It calculates the square root of an incoming 16-bit **signed** number.
- It provides a **done** status signal, and an **err** flag is asserted if the incoming value was negative (no real sqrt).
- Write a testbench (**sqrt_tb.sv**) to test the functionality of this circuit.
- Your testbench should reset the DUT and perform at least 3 different calculations.

→ **Submit sqrt_tb.sv**



Signal	Dir	Description
clk, rst_n	in	clk & active low asynch reset
go	in	When asserted op is sampled and sqrt calculation commences.
op[15:0]	In	signed operand to take sqrt of
sqrt[7:0]	out	result
done	out	Asserts when calculation completed, deasserts with next go
err	out	Last value had no sqrt